
Exercise 1 - Forward networks

Question 1

spirals) Multilayer convolutional networks¹ for nonlinear classification (crescents and

Dataset: Use the provided scripts `py.crescents_2q_make` and `py.spirals_2q_make`.

Only the `seed_run_student` parameter can be changed.

Investigating how to improve nonlinear classification using multilayer architectures.

Objective:

(A) Base model

– Train a Perceptron network with Sigmoid activation on each dataset.

– Report the accuracy of training/validation/testing and show the decision boundary.

Analysis: Explain why the linear model is unable to completely separate the classes.

(b) Nonlinear model

Train with one hidden layer (e.g. 10 neurons, ReLU activation).

– An MLP

And use evaluation.

– From the same error criteria

– Compare the learning curve and decision boundary with the linear baseline.

Give. Analysis: Summarize how the hidden layer changes separability and generalizability.

(c) Study of hyperparameters

– Examine the impact of the following:

$\{10^{-2}, 10^{-3}\}$ 1 - Learning rate with values of $\{10^{-4},$

2 - Hidden layer size with values $\{20, 10, 5\}$

MLP1

– Summarize the mean $\pm$ standard deviation of the test accuracy in the table.

– Draw learning graphs for each experiment.

Analysis: Identify the most stable and accurate setting and explain the observed trends.

(d) Improved *MLP* design (spiral dataset)

– Propose an improved MLP that correctly classifies the spiral dataset.

– You can change the number of layers, neurons, activation type, regularization, optimizer, or learning rate program.

Report:

Number of parameters. – Final architecture summary and

– Training/validation/testing performance, learning graphs and decision boundaries.

– Comparison table: Base model vs. MLP.

Analysis: Explain why your architecture performs better.

Explanations

1 - All code must be implemented with Python and the PyTorch framework.

2 - Reports must be structured and include all graphs, metrics, and interpretations. Code without explanation Score

It doesn't work.

3. Cheating or unauthorized collaboration is prohibited. Violating students will be prosecuted.

4 - The use of external resources or content generated by artificial intelligence for learning is permitted, but the work presented

It must be fully understood and explained by the student.

10% Split Validation 5 - Unless otherwise stated, data should be divided into 70% training, 20% testing, and

Become.

And submit your report before the deadline. Every day 6 - A compressed file named zip01.HW_StudentID containing the code

Give. Tardiness reduces the score by 10%.

7 - For any questions about the training, you can contact the course's teaching assistants via the following email:

ann.ceit.aut@gmail.com